

Adroit Technologies

Protocol Driver

Name	CANBUS Client Protocol Driver (KVaser API)
DLL	D_CANBUS_CL



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1. Overview

CANBUS is partially supported by this driver, additions will be added as required.

Using the KVASER canlib API version 5.49.188, simulator is provided by canking version 6.27.701. CanDrivers version 5.49.188.

Each Adroit Agent Server can enable tags linked to a CANBUS message provided by a browser that will load a dbc file.

Raw 8 bytes , Unsolicited message support, by means of a dbc file that defines one or more CANid messages and one or more signal per message. A CANid range is specified in setup , the driver will then not allocate these to a scantask nor attempt to poll for them in anyway.

1.1. Supported Hardware

This driver has been tested on the following hardware:

Kvaser simulated ports using CANKING.

2. Wiring

Not applicable.

3. Device Configuration

3.1. Device Configuration

CANBUS Client driver - CAN1

☐ Enable secondary
 ☒ Primary
 ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby
 ☒ Enable standby reads
 ☐ Enable standby writes

Property	Value
CANbus details	
Channel index	0
Timeout	3000
DBC file	D:\temp_canbus\Web\dbc\UDS_VW.dbc
Bitrate	250 kbit/s
Use Flexible bitrate	False
Flexible bitrate	1 Mbit/s 80% sampling
Unsolicited mode	
Contact exceeded period (mSec.)	120000
Begin CAN ID (dec.)	1024
End CAN ID (dec.)	1279
Advanced details	
Debug level bitmask	65535
Stale threshold	0

Close

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

3.1.1. Driver Details

Enable Secondary – (Default = OFF) Allows enabling of redundant MQTT Broker connections such that in case of a failure a switchover will occur.

Primary – Allows configuration setting of the Primary broker connection.

Secondary – Allows configuration setting of the Secondary broker connection.

3.1.2. Device Details

Enable shadow scanning from master to standby – (Default = OFF) If enabled will stop any communication to the MQTT broker when the Agent Server is in standby mode. By default, we assume parallel scanning.

Enable standby reads – (Default = ON) Allows reading from the MQTT broker when the Agent Server is in standby mode.

Enable standby writes – (Default = OFF) Allows writing to the MQTT broker when the Agent Server is in standby mode.

CANbus details

Channel Index – Channel index as per defined within the CANbus API. Usually exclusive open, all channel on the bus “hear” the other channels. Can be thought of as nodes.

Timeout – timeout in milliseconds, if polling a specific address. Not applicable for Unsolicited updates , see contact exceeded period below.

DBC file– CSV of the full filename and path, multiple dbcs will be supported, currently it is recommended to only have one active at a time. Contact Adroit Drivers if this is not adequate.

Bitrate – dropdown combo of applicable bitrate for the bus , the driver must match the CANbus hardware.

Use flexible bitrate – opt in to allow for flexible higher than normal rates, the driver must match the CANbus hardware. Only one type of bit rate is active at a time so if this is true then Flexible bitrate is active.

Flexible bitrate – dropdown combo of additional applicable bitrates for the bus , the driver must match the CANbus hardware.

Unsolicited mode

Contact exceeded period (mSec.) Session – (Default = True) This will tell the MQTT broker to establish a persistent session or not. If True, the broker will discard any data previously associated with the current client ID once successfully connected. Otherwise, when disconnected the broker and client will store the session, all subscriptions for the client and all missed messages, when QoS 1 or 2.

Begin CAN ID (dec) – (Default = 1024dec) if the CANID is between this begin and end value this item will not be allocated to a read task , but only available for unsolicited updates, the driver will listen on the bus for this item.

End CAN ID (dec) – (Default = 1279dec) if the CANID is between this begin and end value this item will not be allocated to a read task , but only available for unsolicited updates, the driver will listen on the bus for this item.

Advanced details

Debug level bitmask – (Default = 65535) This is a bitmask that enable extra debugging in the datascopes output, do not leave it enabled during normal operation on high-speed drivers as it will slow down the system depending on speed requirements. In serial drivers this delay is negligible.



INFORMATION

Remember, all NASA drivers one can stop and start the device to reload these flags.

960 default is just the protocol messages 40hex + 80hex+ 100hex + 200hex = 3C0hex -> 960 decimal in the dialogue.

#define DEBUG_MASK_COMSRESOURCE_INFO	0x00000001
#define DEBUG_MASK_COMSRESOURCE_CRITICAL	0x00000002
#define DEBUG_MASK_PROTOCOL_ITEM_VALUES	0x00000004
#define DEBUG_MASK_PROTOCOL_ITEM_BAD_VALUES	0x00000008
#define DEBUG_MASK_NASA_POOLS	0x00000010
#define DEBUG_MASK_TIMING	0x00000020
#define DEBUG_MASK_PROTOCOL_RESPONSES	0x00000040
#define DEBUG_MASK_PROTOCOL_REQUESTS	0x00000080
#define DEBUG_MASK_PROTOCOL_READ_DESC	0x00000100
#define DEBUG_MASK_PROTOCOL_WRITE_DESC	0x00000200
#define DEBUG_MASK_PROTOCOL_APILOG	0x00000400

Stale threshold – (Default = 0) Every nth occurrence of an unchanged item value will be sent to the master. If the value is different, the value will be sent regardless of this count. This count reduces unnecessary events within the Agent Server.

4. Supported Addresses



PLEASE NOTE

Only REAL or ANALOGUE or INTEGER agent types are supported at this time.

The scan addresses in CANBUS are complex, please use the browser , the setup dialog also provides a view into a loaded database file (dbc).

The dbc loader may adjust the signal notation so as to assist the identification within the driver operation so the view might be slightly altered.

CANBUS Client Address Browser

D:\temp_canbus\Web\dbc\UDS_VW.dbc

Add dbc Remove dbc OK Cancel

Connect

Message: PID_F40C_Engine_RPM_Engine_RPM, CANId=0x400, dlc=2
 PID_F40C:400.0:EnginRPM, start=0, len=16, enc=Intel, rep=2, factor=0.250000, offset=0.000000
 Message: PID_F411_Absolute_Throttle_Position_Absolute_Throttle_Position, CANId=0x401, dlc=1
 PID_F411:401.0:AbsolThrotPosit, start=0, len=8, enc=Intel, rep=2, factor=0.392157, offset=0.000000
 Message: PID_F447_Absolute_Throttle_Position_B_Absolute_Throttle_Position_B, CANId=0x402, dlc=1
 PID_F447:402.0:AbsolThrotPositB, start=0, len=8, enc=Intel, rep=2, factor=0.392157, offset=0.000000
 Message: PID_1026_Accelerator_pedal_position_Prc_SWORD, CANId=0x403, dlc=2
 PID_1026:403.0:AccelPedalPosit_PrcSWORD, start=0, len=16, enc=Intel, rep=1, factor=0.012207, offset=0.000000
 Message: PID_F449_Accelerator_Pedal_Position_D_Accelerator_Pedal_Position_D, CANId=0x404, dlc=1
 PID_F449:404.0:AccelPedalPositD, start=0, len=8, enc=Intel, rep=2, factor=0.392157, offset=0.000000
 Message: PID_F44A_Accelerator_Pedal_Position_E_Accelerator_Pedal_Position_E, CANId=0x405, dlc=1
 PID_F44A:405.0:AccelPedalPositE_1, start=0, len=8, enc=Intel, rep=2, factor=0.392157, offset=0.000000
 Message: PID_10E2_Ignition_advance_reduction_cylinder_1_Kw_q0p75_SBYTE, CANId=0x406, dlc=1
 PID_10E2:406.0:IgnitAdvanReducCylin1_KwQ0p75SBYTE, start=0, len=8, enc=Intel, rep=1, factor=0.750000, offset=0.000000
 Message: PID_10E5_Ignition_advance_reduction_cylinder_2_Kw_q0p75_SBYTE, CANId=0x407, dlc=1
 PID_10E5:407.0:IgnitAdvanReducCylin2_KwQ0p75SBYTE, start=0, len=8, enc=Intel, rep=1, factor=0.750000, offset=0.000000
 Message: PID_10E3_Ignition_advance_reduction_cylinder_3_Kw_q0p75_SBYTE, CANId=0x408, dlc=1
 PID_10E3:408.0:IgnitAdvanReducCylin3_KwQ0p75SBYTE, start=0, len=8, enc=Intel, rep=1, factor=0.750000, offset=0.000000
 Message: PID_10E4_Ignition_advance_reduction_cylinder_4_Kw_q0p75_SBYTE, CANId=0x409, dlc=1

Tag scanning configuration : CAN1

Device: CAN1 ☒ Started ☐ Stopped

Agent: _SAMPLEANALOGUE2 ... Find

Slot: ☐ Header slots
 rawValue

Address: PID_F447:402.0:AbsolThrotPositB ... Find

Scan rate (ms): 4000 Deadband: 0

☐ Input disabled ☐ Output enabled
☐ Demand scan ☐ Output initialize

Currently scanned tags: Scan Unscan

Tag	Address	Rate	Deadband	IO flag
_SAMPLEANALOGUE....	PID_F40C:400.0:EnginRPM	4000	0	1
_SAMPLEANALOGUE...	PID_F447:402.0:AbsolThrotPositB	4000	0	1

For example:

5. Protocol Definition and Considerations

```
ASCII DATASCOPE for CAN1 on Pri,(A) Channel 0 *LOG

Updating Adroit
Addr= PID_F40C:400.0:EnginRPM, Val= 3.75, PrevVal= 5, Q=Good, QOld=192, QChanged=0

Updating Adroit
Addr= PID_F40C:400.0:EnginRPM, Val= 3.75, PrevVal= 5, Q=Good, QOld=192, QChanged=0

Idle for 30016, State engine step : 3
```

```
ASCII DATASCOPE for CAN1 on Pri,(A) Channel 0 *LOG

Idle < Stopped >...
Idle < Stopped >...

CANBUS_Device Started
<Stopping> refreshing configuration from registry
Canbus canBusOff -> Successful, Handle:0, ChannelIndex:0
Canbus canClose -> Successful, Handle:0, ChannelIndex:0
API requesting Disconnect() SUCCESS [ConnectCount:-1]
<Starting> refresh config(s) from registry
API requesting Disconnect() SUCCESS [ConnectCount:-2]

Stand-alone server detected

In Open()
OpenWork - API Initialize()...

TIMEOUT <mSec> going to -> 3000

Canbus library initialised
- DBC file already loaded -> D:\temp_canbus\Web\dbc\UDS_UW.dbc , skipping load

Canbus library LoadDB <D:\temp_canbus\Web\dbc\UDS_UW.dbc> -> Successful
Canbus canOpenChannel -> Successful, Handle:0, ChannelIndex:0
Canbus setBusParams -> Successful, Handle:0, ChannelIndex:0
Canbus canBusOn -> Successful, Handle:0, ChannelIndex:0
OpenWork - CANBus Open() SUCCESS [ConnectCount:-1]

Idle for -1341207083887, State engine step : 3

<PAUSED>
```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroitscada.com

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Sparkplug B Notes

For Adroit 10.0.5.5 onwards Sparkplug B is supported via a SmartUI Server script.

That usually scans for all expected messaging.

Input and output Sparkplug B messages are usually scanned to a relatively high-level wildcard "K:#" This will allow all Sparkplug messages to be sent to the applicable slots usually a String Agent's valueex slot.

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	15-Jan-2026	D Jordaan	First revision of this doc.	D. Jordaan
1.1				
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
1	15/01/2026	First release of the driver. Unsolicited support for Raw signals